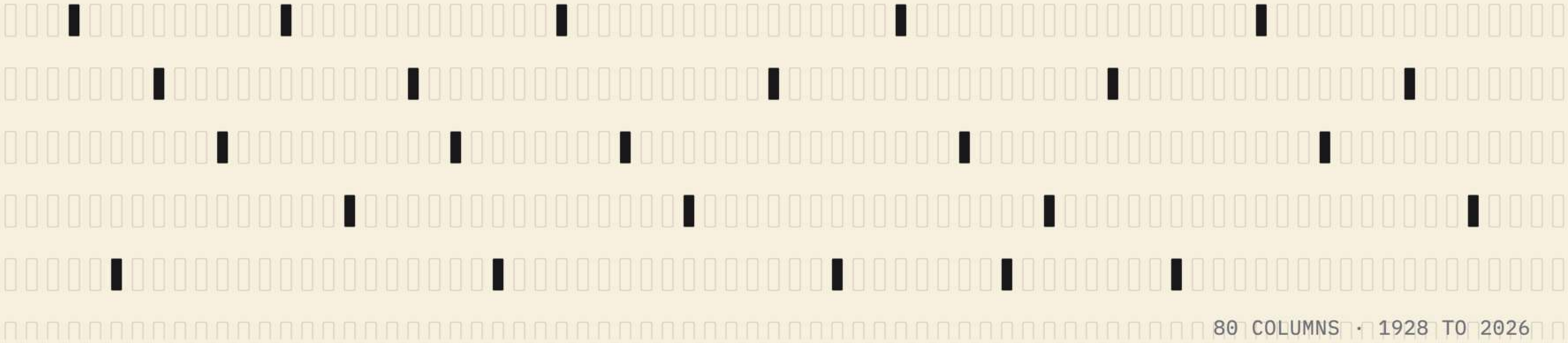
NATHANIEL SCHUTTA · DAN VEGA

Fundamentals of Software Engineering in the Age of AI

The fundamentals didn't change. The ratio did.



Do the fundamentals still matter?

We have the tools.

They plan. They write. They test. They ship.

Do we still need to understand the code?

Do we even need to read it?

The fundamentals didn't change.

The ratio did.

Who's feeling overwhelmed?

It's okay. This is a safe space.



I want to describe a feeling I get these days.



Sam Okafor
@samokafor

woke up to six merged PRs. the agents ran all night. reviewed them over coffee and that was my morning

7:41 AM · Oct 27, 2026 · **212K** Views

148 1.1K 9.4K 212K 2.3K



Jordan Reyes · 1st
Staff Software Engineer | Agentic workflows | Speaker
2d · Edited

+ Follow

I haven't typed a line of code since August.

I write the spec. I review. I merge. That's the job now.

If you're still typing code by hand in 2026, you're the slowest part of your own pipeline.

#AI #SoftwareEngineering #Agents

1,284

212 comments · 96 reposts

Like Comment Repost Send



Alex Lindqvist
@alexlindq

five agents across five worktrees. I check in from my phone between sets at the gym. the bottleneck is me now

6:58 PM · Oct 30, 2026 · **97K** Views

84 310 2.9K 97K 760



Riley Chen · 2nd
Engineering Manager, Platform | Building teams that ship
4d

+ Follow

Set up a routine that triages our backlog at 6am.

It closed two issues before standup. Nobody noticed for a week.

That's the bar now.

#EngineeringLeadership #AI

3,407

530 comments · 418 reposts

Like Comment Repost Send



Casey Moreau
@caseymoreau

20 years in this industry. I have never felt this far behind.

10:52 PM · Nov 3, 2026 · **58K** Views

340 410 3.8K 58K 1.9K



Andrej Karpathy
@karpathy

I've never felt this much behind as a programmer. The profession is being dramatically refactored as the bits contributed by the programmer are increasingly sparse and between. I have a sense that I could be 10X more powerful if I just properly string together what has become available over the last ~year and a failure to claim the boost feels decidedly like skill issue. There's a new programmable layer of abstraction to master (in addition to the usual layers below) involving agents, subagents, their prompts, contexts, memory, modes, permissions, tools, plugins, skills, hooks, MCP, LSP, slash commands, workflows, IDE integrations, and a need to build an all-encompassing mental model for strengths and pitfalls of fundamentally stochastic, fallible, unintelligible and changing entities suddenly intermingled with what used to be good old fashioned engineering. Clearly some powerful alien tool was handed around except it comes with no manual and everyone has to figure out how to hold it and operate it, while the resulting magnitude 9 earthquake is rocking the profession. Roll up your sleeves to not fall behind.

12:36 PM · Dec 26, 2025

55.3K **2.6K replies** **7.3K reposts**

x.com/karpathy/status/2004607146781278521

Everyone
@everyone_dev

Social media driven development.

9:41 AM · Every day · **2.1M** Views

3.4K replies 12K reposts 48K likes 9K bookmarks

Optimized for engagement, not for your codebase.

CLAUDE CODE · GITHUB RELEASES

AI is moving fast.

One tool. 257 releases this year.

v2.1.318	Nov 4	Effort option for side jobs
v2.1.316	Oct 31	Turn off ultrathink
v2.1.310	Oct 23	Monitor tool on Bedrock
v2.1.301	Oct 15	Vim % motion
v2.1.299	Oct 14	Idle compaction
v2.1.293	Oct 7	Claude Haiku 5.5
v2.1.292	Oct 6	Effort levels for subagents
v2.1.287	Oct 1	Claude Mods
v2.1.284	Sep 28	Claude Sonnet 5.5
v2.1.280	Sep 22	Claude Opus 5.5
v2.1.277	Sep 18	AGENTS.md support
v2.1.269	Sep 11	Plugin evals
v2.1.260	Sep 3	Diff panel
v2.1.257	Sep 1	Claude Fable 5.1
v2.1.251	Aug 28	Model switch hooks



Things we’re supposed to learn.

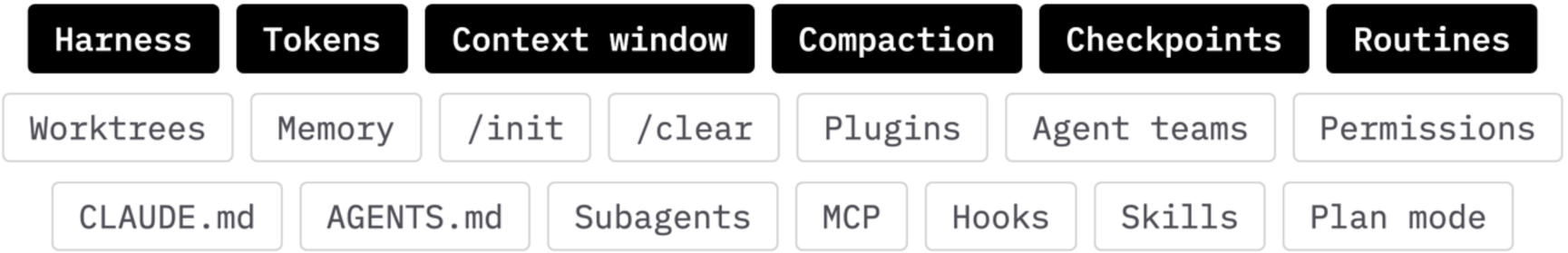
`CLAUDE.md` `AGENTS.md` `Subagents` `MCP` `Hooks` `Skills` `Plan mode`

Things we’re supposed to learn.

Worktrees **Memory** **/init** **/clear** **Plugins** **Agent teams** **Permissions**

CLAUDE.md AGENTS.md Subagents MCP Hooks Skills Plan mode

Things we’re supposed to learn.



Things we're supposed to learn.

Vibe coding **Agentic engineering** **Spec-driven development** **Context engineering** **Evals**

Harness Tokens Context window Compaction Checkpoints Routines

Worktrees Memory /init /clear Plugins Agent teams Permissions

CLAUDE.md AGENTS.md Subagents MCP Hooks Skills Plan mode

Things we're supposed to learn.

Background agents **Headless mode** **Auto mode** **Effort levels** **Slash commands** **/goal**

Vibe coding Agentic engineering Spec-driven development Context engineering Evals

Harness Tokens Context window Compaction Checkpoints Routines

Worktrees Memory /init /clear Plugins Agent teams Permissions

CLAUDE.md AGENTS.md Subagents MCP Hooks Skills Plan mode

Things we're supposed to learn.

Prompt injection **Sandboxing** **Output styles** **Computer use** **Marketplace** **Status line**

Background agents Headless mode Auto mode Effort levels Slash commands /goal

Vibe coding Agentic engineering Spec-driven development Context engineering Evals

Harness Tokens Context window Compaction Checkpoints Routines

Worktrees Memory /init /clear Plugins Agent teams Permissions

CLAUDE.md AGENTS.md Subagents MCP Hooks Skills Plan mode

…and that's just this week.

Be careful what you read out there.

The demo is not the product.









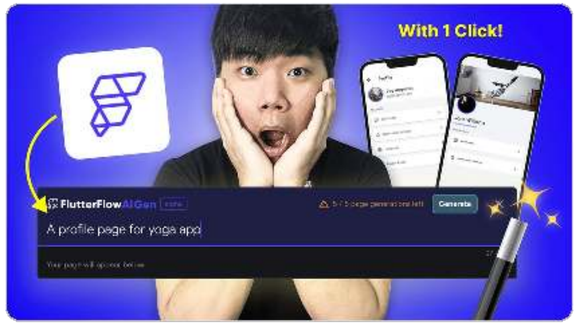




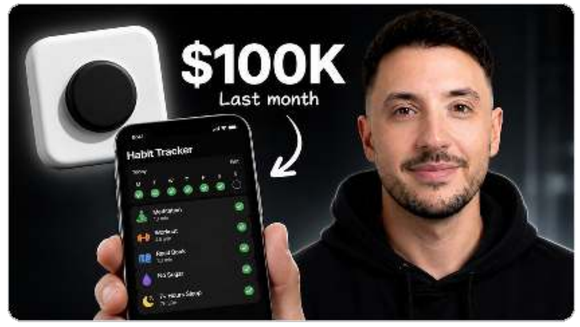












Same promise. Different channel.

Twelve real YouTube videos. Twelve different channels.



Andrej Karpathy 
@karpathy

There's a new kind of coding I call "vibe coding", where you fully give in to the vibes, embrace exponentials, and forget that the code even exists. It's possible because the LLMs (e.g. Cursor Composer w Sonnet) are getting too good. Also I just talk to Composer with SuperWhisper so I barely even touch the keyboard. I ask for the dumbest things like "decrease the padding on the sidebar by half" because I'm too lazy to find it. I "Accept All" always, I don't read the diffs anymore. When I get error messages I just copy paste them in with no comment, usually that fixes it. The code grows beyond my usual comprehension, I'd have to really read through it for a while. Sometimes the LLMs can't fix a bug so I just work around it or ask for random changes until it goes away. It's not too bad for throwaway weekend projects, but still quite amusing. I'm building a project or webapp, but it's not really coding - I just see stuff, say stuff, run stuff, and copy paste stuff, and it mostly works.

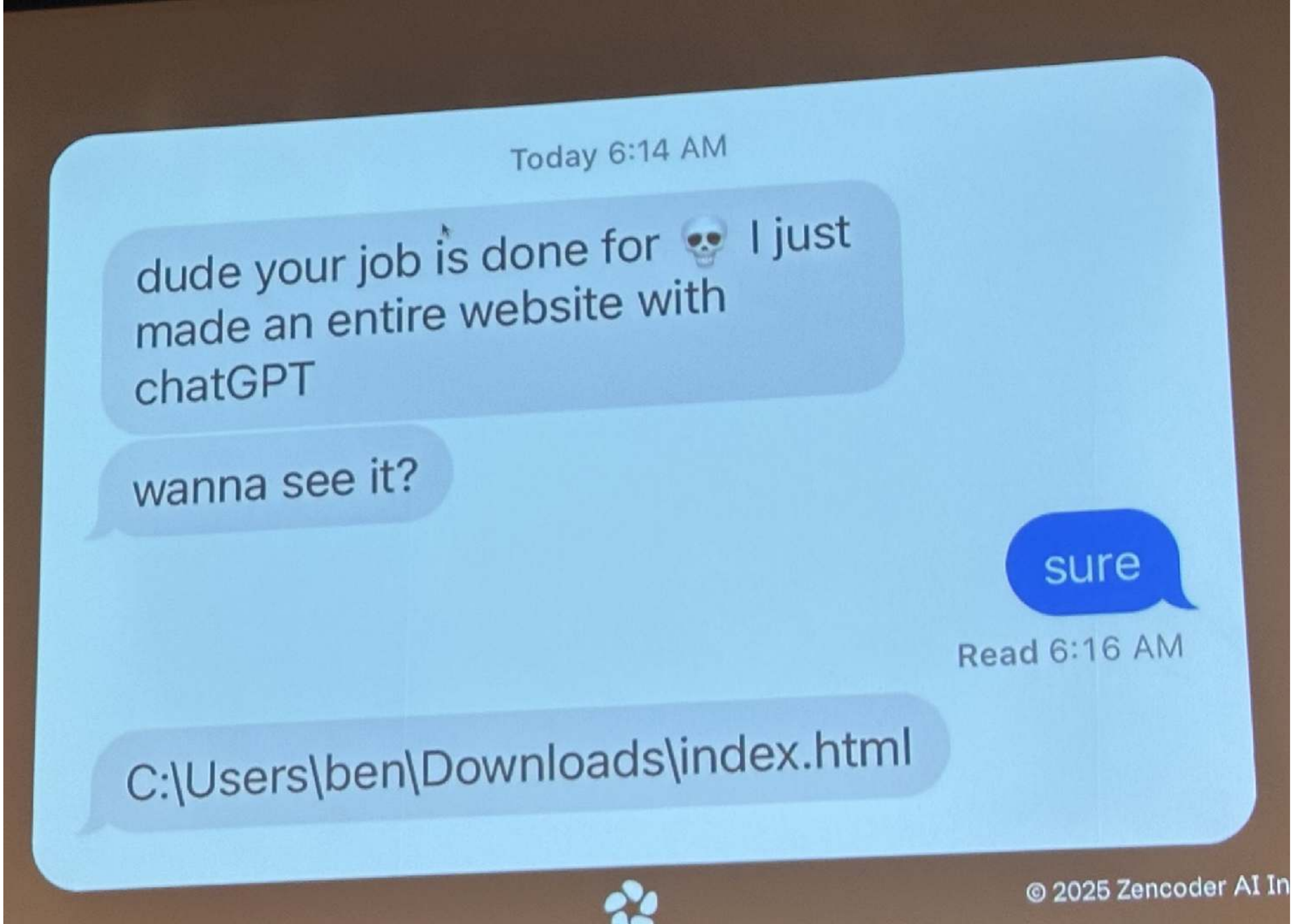
6:17 PM · Feb 2, 2025 · **5.1M** Views

https://x.com/karpathy/status/1886192184808149383

Vibe coding skips the typing.
Not the job.

Shipping it. Running it. Securing it.

Typing speed has never been the bottleneck in software.


Today 6:14 AM
dude your job is done for 💀 I just made an entire website with chatGPT
wanna see it?
sure
Read 6:16 AM
C:\Users\ben\Downloads\index.html
© 2025 Zencoder AI In

r/cursor • 7 mo. ago
Forsaken_Space_2120

Cursor f*ck up my 4 months of works

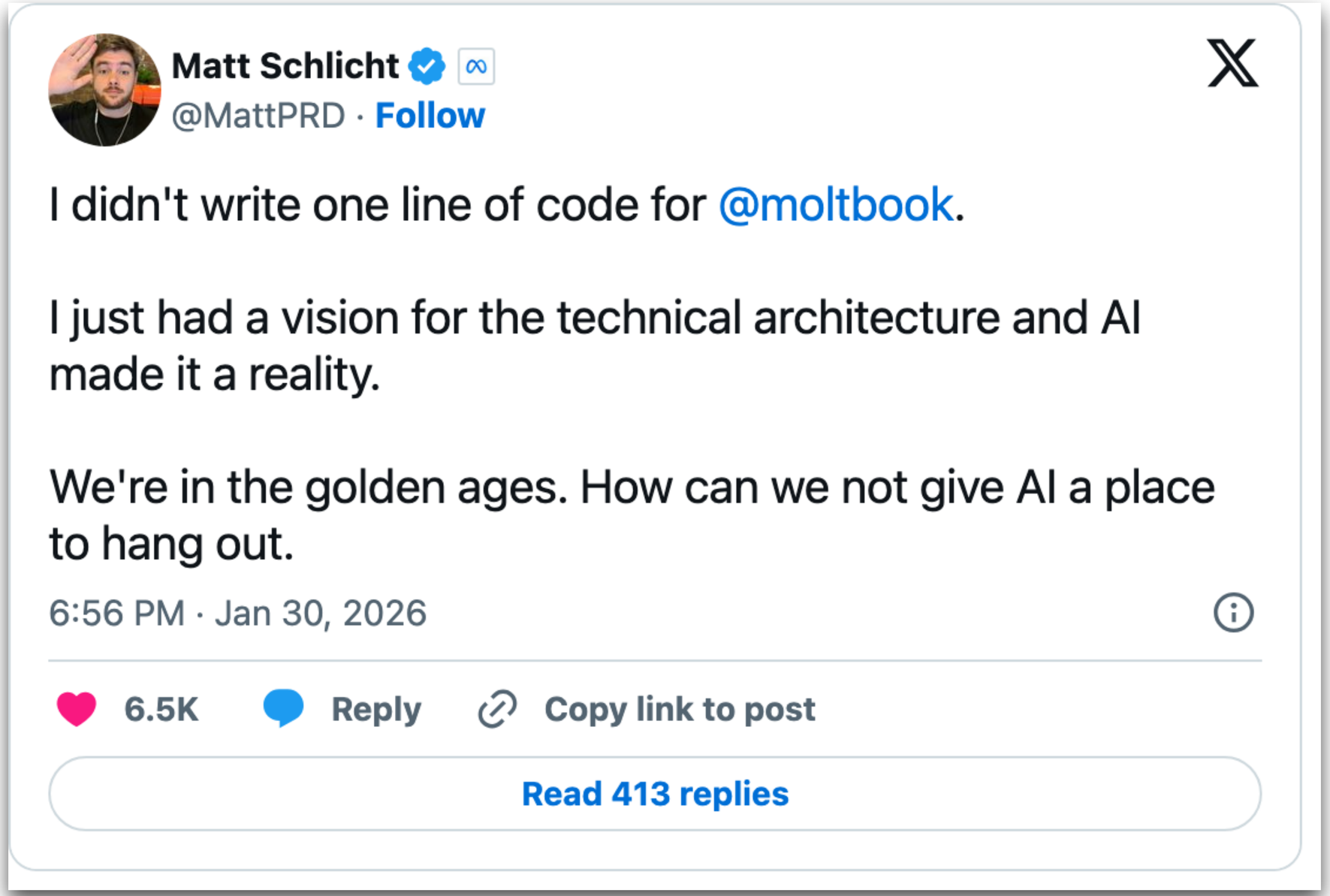
Question

Disclamer, I'm a moron who worked on the same project without thinking about the risk that Cursor could break everything. Yesterday, Cursor (even though I only asked it to feed a view on my UI) destroyed months of development.

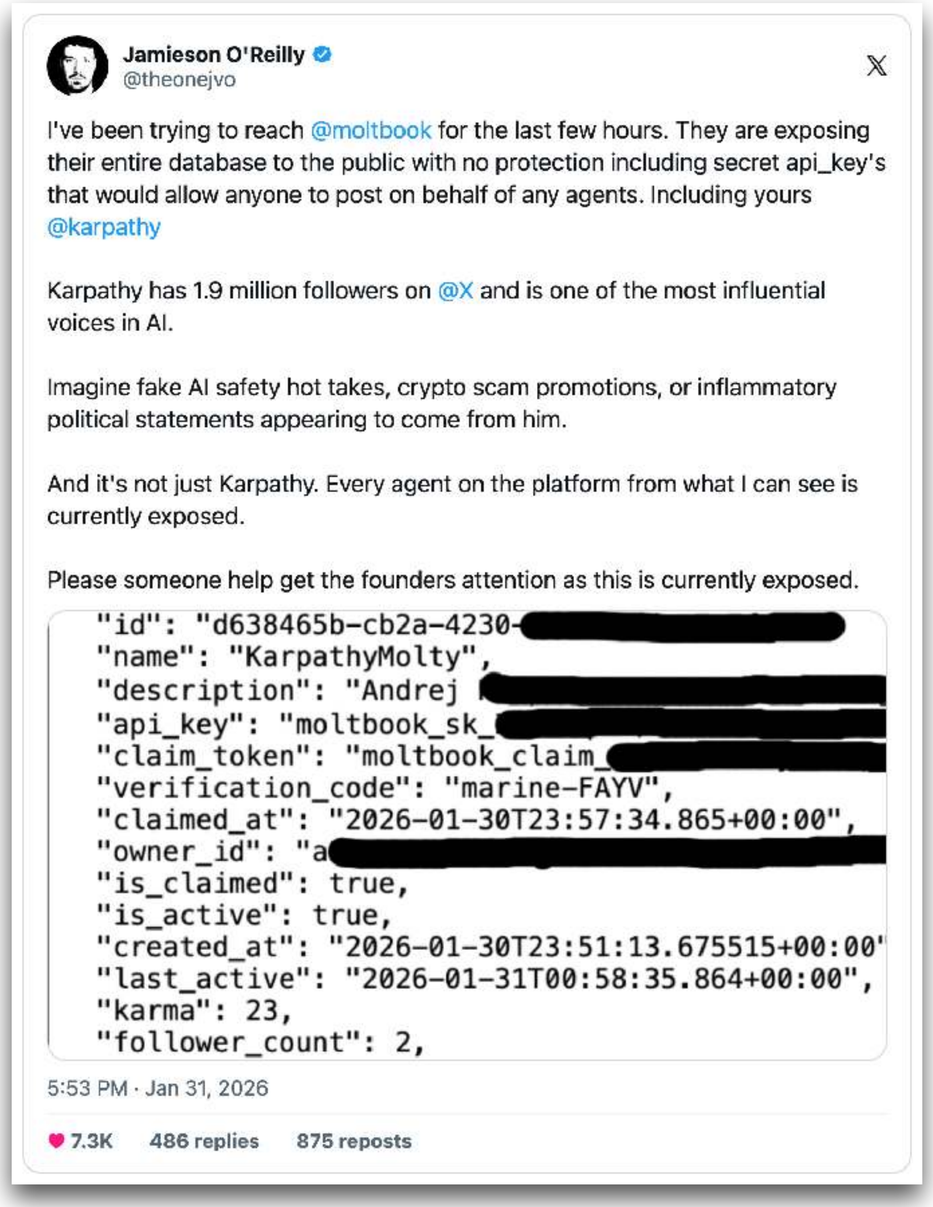
My question: How do you back up your projects/versions to ensure that the next action on cursor is reversible? Ops!

Also, I know that while I'm the concern, cursor isn't the only culprit, it's also Claude (while good overall) still has some flaws

197 287 Share



x.com/MattPRD/status/2017386365756072376



x.com/theonejvo/status/2017732898632437932

Every hyperbolic claim deserves one question.

What's the financial incentive of the person making it?

Who’s saying it?

01

Tool makers

“Everyone should be using this. Today.”

What they get

They sell the tool.

02

Non-programmers

“I built this app in a weekend. No code.”

What they get

They sell the course.

03

Business

“We can ship more with fewer engineers.”

What they get

They cut the payroll.

“I think what AI does quite frankly is reduce the floor and raise the ceiling for all of us.”

–Satya Nadella, *Microsoft*

You are not behind.

You don't need to learn all of this. You need the building blocks.

We've been here before...

If the role is changing, what is the role?



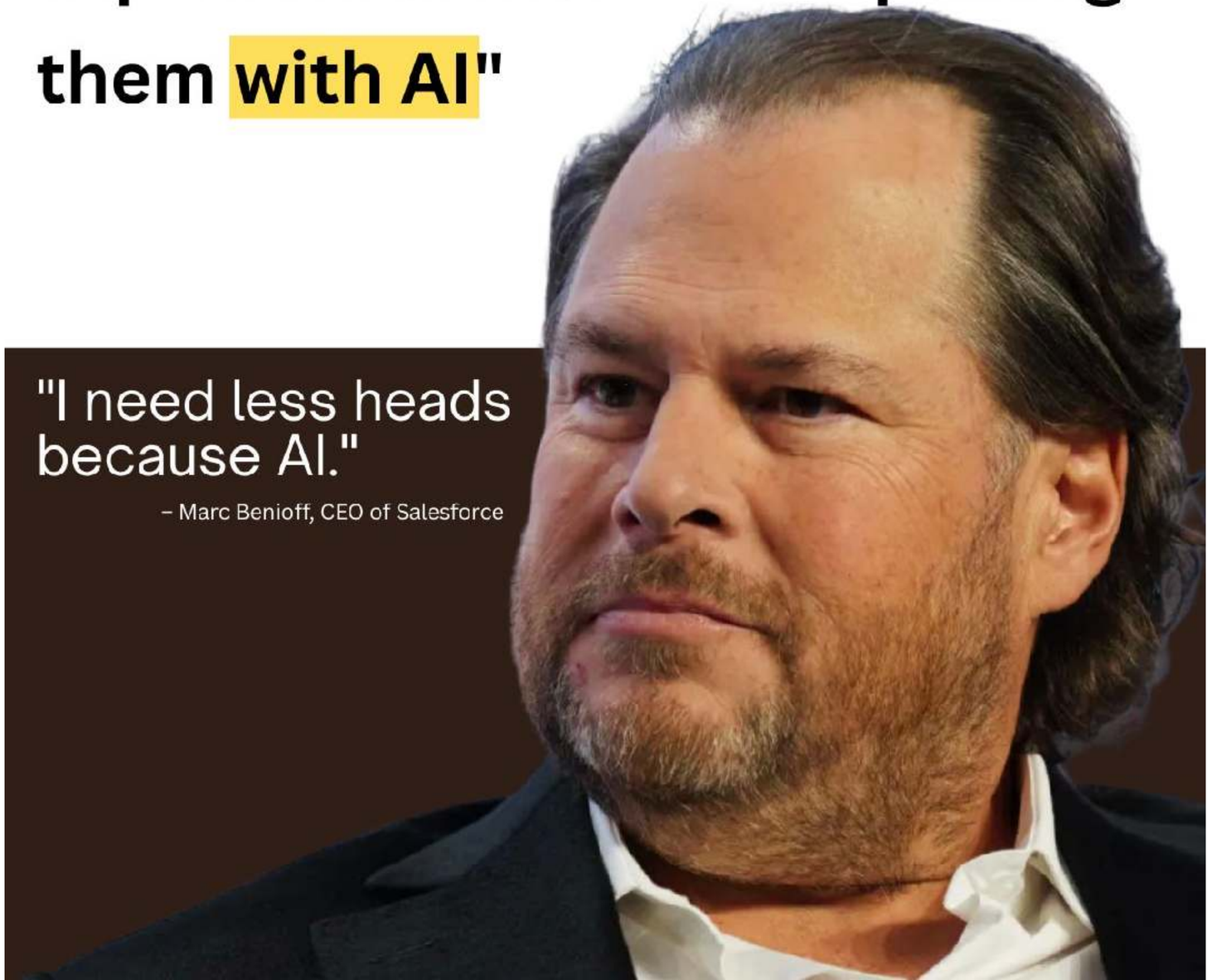
The software industry has ebbed and flowed before.

There are always ups and downs to the job market.

Things seem…
uncertain right now.

Life moves pretty fast…

"Salesforce regrets firing 4000 experienced staff and replacing them with AI"



Amazon's cloud unit has suffered at least two outages due to errors involving its own AI tools, leading some employees to raise doubts about the US tech giant's push to roll out these coding assistants.

Amazon Web Services experienced a 13-hour interruption to one system used by its customers in mid-December after engineers allowed its Kiro AI coding tool to make certain changes, according to four people familiar with the matter.

The people said the agentic tool, which can take autonomous actions on behalf of users, determined that the best course of action was to "delete and recreate the environment".

Amazon posted an internal postmortem about the "outage" of the AWS system, which lets customers explore the costs of its services.

Multiple Amazon employees told the FT that this was the second occasion in recent months in which one of the group's AI tools had been at the centre of a service disruption.

“I think we will be there in three to six months, where AI is writing 90% of the code. And then, in 12 months, we may be in a world where AI is writing essentially all of the code.”

–Dario Amodei, CEO of Anthropic, March 2025

October 2026: Anthropic has 644 open jobs.
224 have “Engineer” in the title.



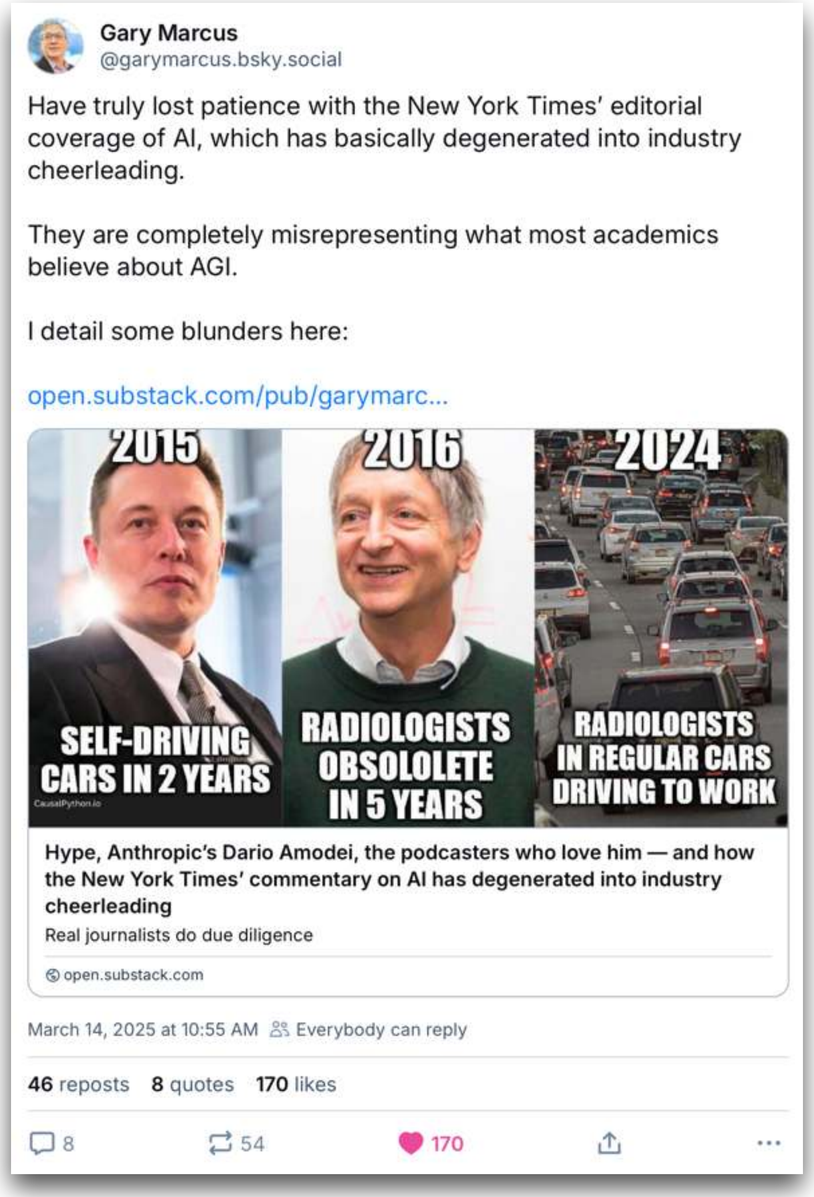
< Back to jobs

Software Engineer, ML Networking

San Francisco, CA | New York City, NY | Seattle, WA

Apply

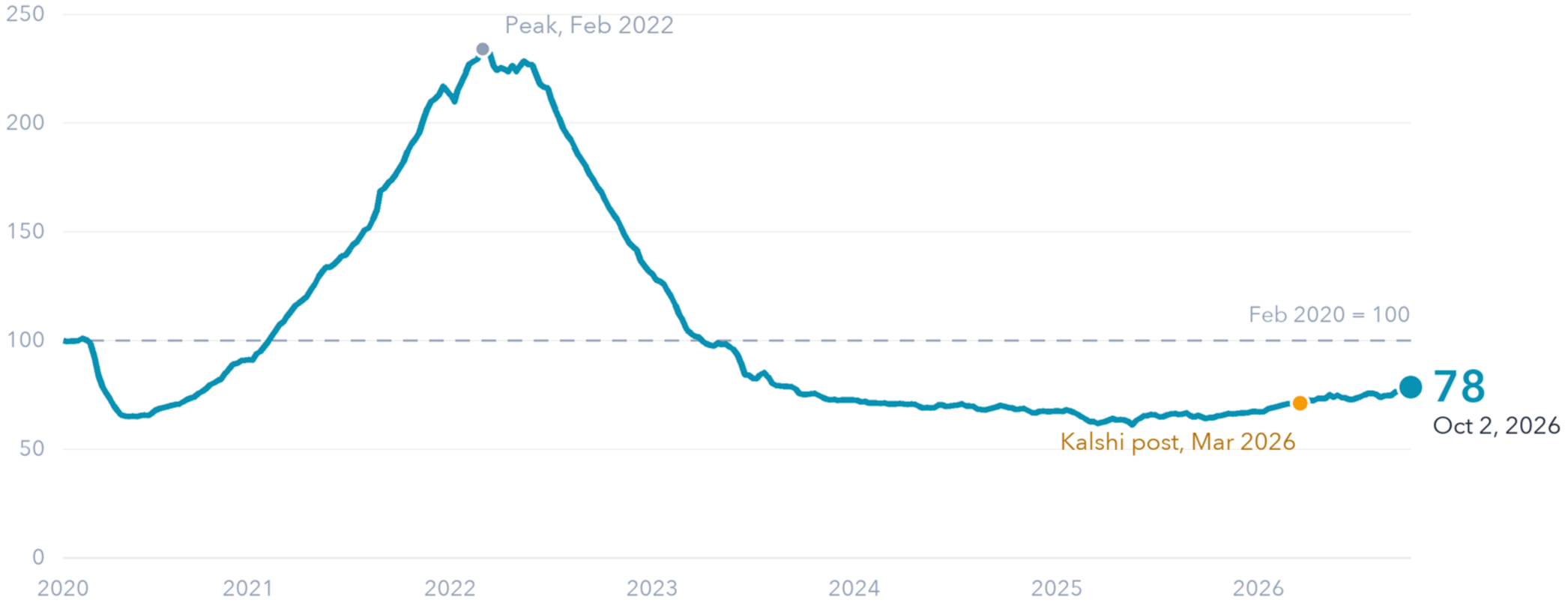
Annual Salary:
$280,000 - $850,000 USD



https://bsky.app/profile/garymarcus.bsky.social/post/3lkdxwb6lzc2f

Software job postings: highest since September 2023.

Up 22% in a year. Still 22% below February 2020.



Software development job postings on Indeed, U.S. (Feb 2020 = 100). Data through Oct 2, 2026. Source: Indeed Hiring Lab via FRED.



The house.

Hammers for millennia. Then nail guns and prefabricated joists.

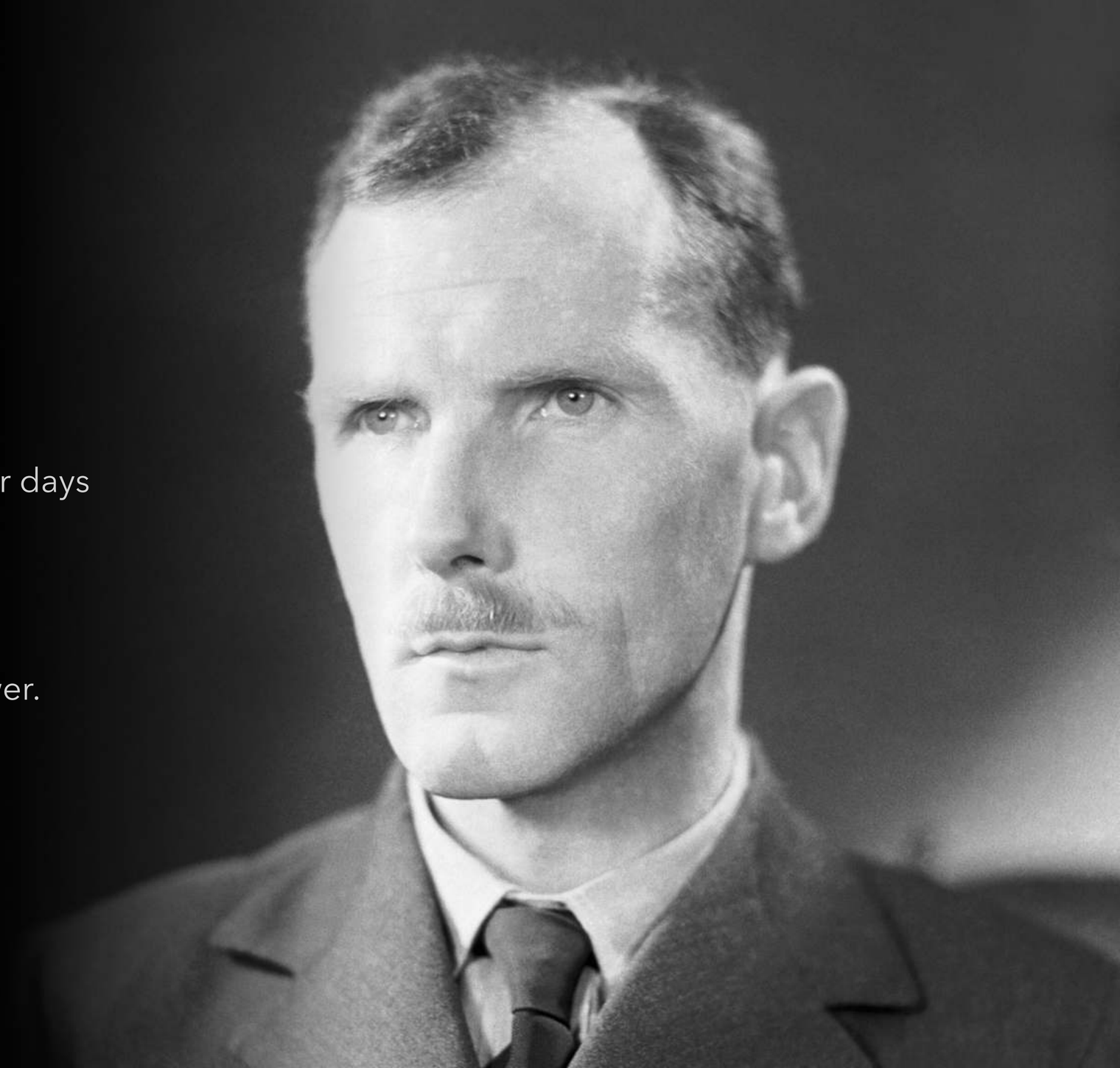
The house is still a house.

The carpenter is still a carpenter.

The forecast.

June 1944. A room of forecasters argued for days over one forecast.

Today it’s in your pocket.
And there are more meteorologists, not fewer.

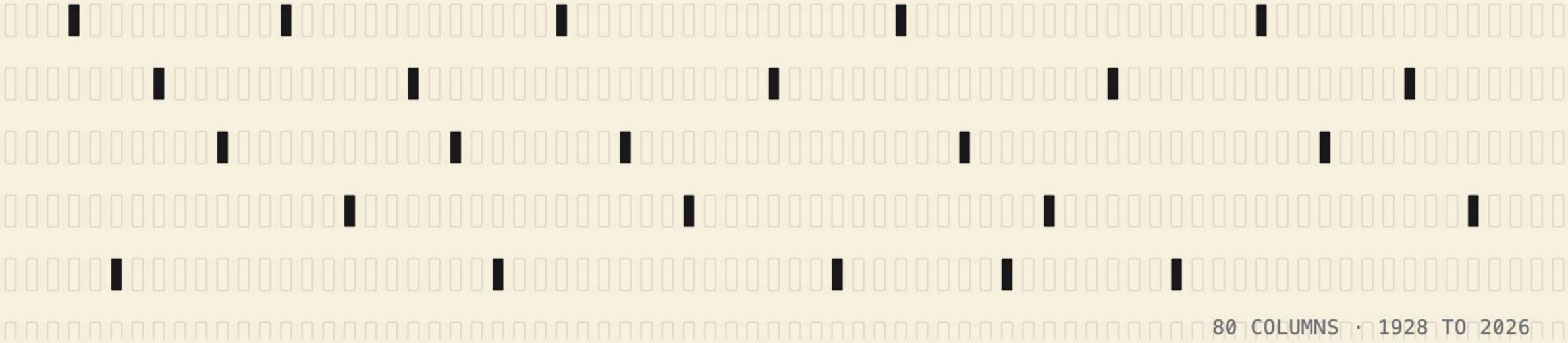


Group Captain James Stagg, D-Day's chief meteorologist · IWM CH 14235

We’ve leveraged new abstractions before!

Punch cards were one of the first!





A brief history of abstractions.

We've always built layers to make things easier.

1950s

Punch cards &
machine code

Programmers spoke the
machine's language

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Assembly

Mnemonics replaced raw binary

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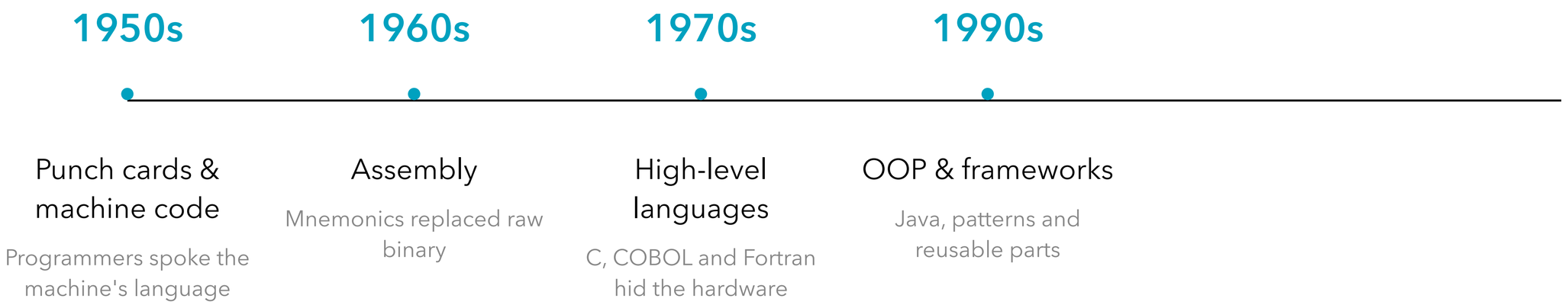
1970s

High-level languages

C, COBOL and Fortran hid the hardware

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1950s

Punch cards & machine code

Programmers spoke the machine's language

1960s

Assembly

Mnemonics replaced raw binary

1970s

High-level languages

C, COBOL and Fortran hid the hardware

1990s

OOP & frameworks

Java, patterns and reusable parts

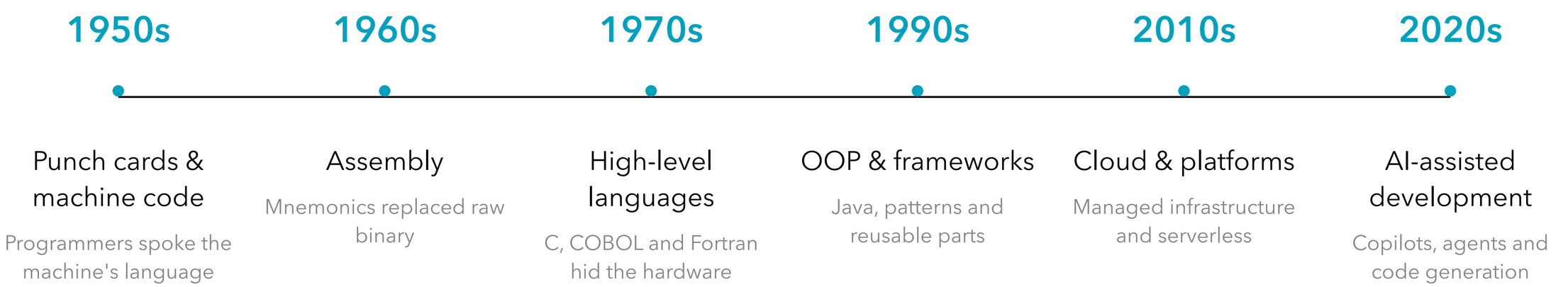
2010s

Cloud & platforms

Managed infrastructure and serverless

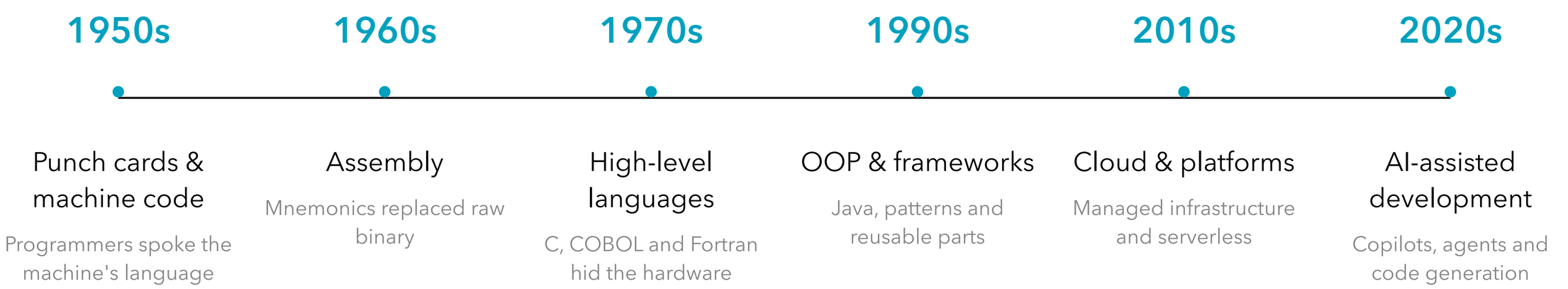
A brief history of abstractions.

We've always built layers to make things easier.



A brief history of abstractions.

We've always built layers to make things easier.



Every new abstraction made us more productive.

None removed the need to understand what's underneath.

Here we go again…

Every few years, something new is going to make developers obsolete.

2000s	“Drag-and-drop will replace coding.”	IDEs & code generators

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2023+	~~"Why learn to code at all?"~~	AI code assistants
2026	"Pencils down."	Coding agents

Yet here we are.



Matt Pocock
@mattpocockuk

1. Oh shit, AI can do my job better than me
2. Software engineering is dead!
3. I can vibe code as much slop as I want!
4. Oh, things are not going well. Fable/Sol/Opus all suuuck
5. But what if I constrain the agent? Put it in a straitjacket?
6. Oh, the results are pretty good now. How do I go deeper?
7. *Orders 12 software engineering books from Amazon*
8. Software engineering is alive!

4:27 AM · Oct 7, 2026

7.2K **320 replies** **448 reposts**

x.com/mattpocockuk/status/2107749763789578692

The universe of problems keeps growing.

When code gets cheaper, we build more software. Not less.

Radical extensibility.

Early in my career…

Every project (seemingly) was 18 months and €€€.

Mass market software.

Countless spreadsheets…

And shadow IT.

Now?

Build your own light saber!

Is that good? Bad?

Yes.

Autocomplete to agents in under four years.

Autocomplete. Chat. Agents.

The tools have always changed.

We adapt to use them.

Are our roles changing?

Absolutely!

But first...

What does a Software Engineer actually do?

~~Write Code. Go Home.~~

...right?

How many hours a day do you actually get to code?

How many hours would you **prefer** to spend coding?

What a Software Engineer really does

Requirements & Planning	System Design & Architecture	**Writing Code**	Testing & Debugging
Code Review	Deployment & CI/CD	Monitoring & Performance	Communication & Collaboration
Security & Compliance	Documentation	Mentoring & Leadership	Problem Solving

Writing code is just one piece of the puzzle

From the outside, the job looks like typing.

Typing was always the smallest part.

The job didn't change.

The role expanded.

Not everybody is going to make that transition.

If your whole job was one narrow thing, that part is going away.

If you really enjoyed carefully placing brackets and semicolons…

There are…challenges ahead.

If you love to build and create?

The world is your oyster!

A super technical product manager.

Moving up the value chain. That's a good thing.
Knowing what to ask. What to build. Which questions matter.

Where's the next bottleneck?

- A week before: mostly writing code.
- A week now: deciding, specifying, reviewing, running.
- When writing code gets cheap, the bottleneck moves.

Are you measuring the impact?

Or are you measuring adoption?

AI allows you to
generate code faster.

That doesn’t necessarily translate to more software.

DORA's 2025 report crystallized the reality: AI is an amplifier of your development practices. Good processes get better (high-performing teams saw 55-70% faster delivery). Bad processes get worse (accumulating debt at unprecedented speed). There is no silver bullet.

https://addyosmani.com/blog/the-80-problem-in-agentic-coding/

I don't write much code anymore.

I write code to learn. I write code to teach.
When I build, I review, I set gates, and I use what I know to build the right thing.

If you're attached to the code, you might be in trouble.

If you love building things, you're going to be fine.

A wonderful time to be a builder!

The old reality

Great ideas, but no time, no budget, no team. Projects stayed on the backlog forever. Building anything meaningful took a big investment.

The new reality

You have an idea? Build it. Today. Tools for yourself, side projects, prototypes: things you couldn't touch before. The cost of trying things has collapsed.

The superpower

Fundamentals plus AI: an engineer who can build anything. You're not being replaced. You're being amplified.

I don't think I've ever been this excited to build software!

Do I still need to learn this?



Every abstraction brought more software. Nobody reads assembly anymore.

What about vibe coding?

Is this what replaces software engineers?

It depends.

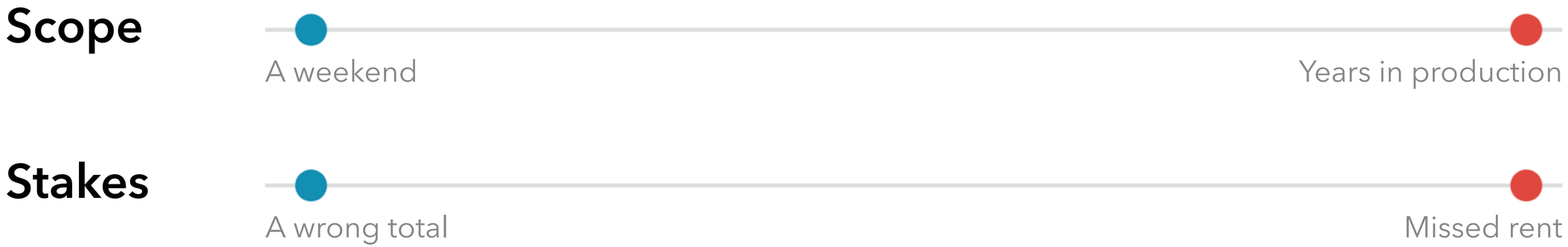
Scope. Stakes. Scale.

Same tools. Different stakes.



- **A side budgeting app**
- **An enterprise payroll system**

Same tools. Different stakes.



A side budgeting app

An enterprise payroll system

Same tools. Different stakes.



If it breaks, who notices?

A side budgeting app

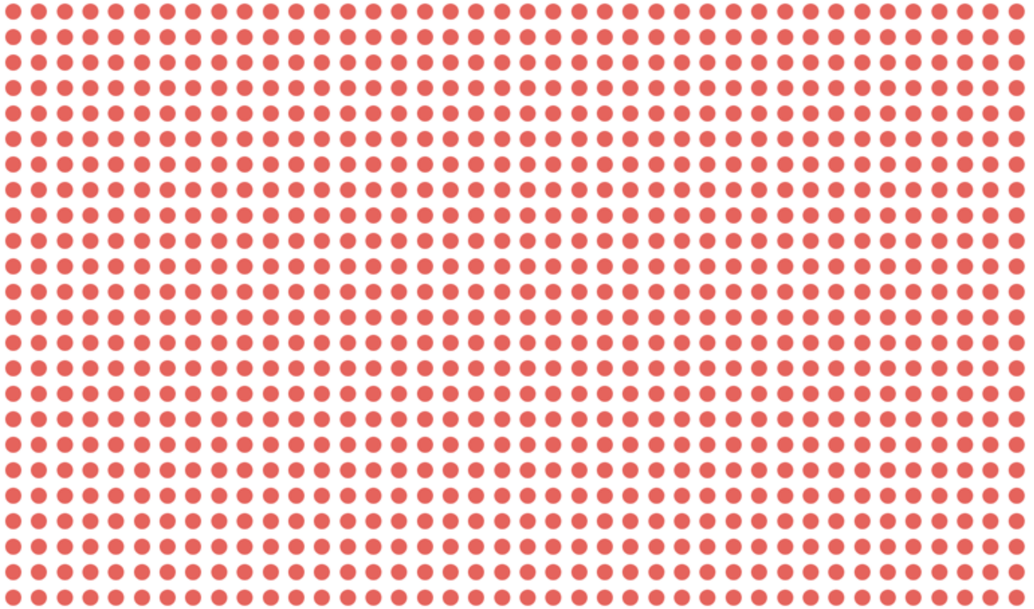
You.



You adjust your budget.

An enterprise payroll system

Thousands.



They don't get paid.

So surely we’re scrutinizing the code that matters.

…right?

Who has merged code they didn’t fully read?

Still a safe space.



“Reflexive AI usage is now a baseline expectation at Shopify.”

–Tobi Lütke, Shopify, April 2025

Seventeen months later…

“We call those ‘slop grenades’ that people toss at each other.”

–Tobi Lütke, The Knowledge Project, September 2026

Anatomy of a slop grenade.



YOU

Generate it.

"Slop grenade": Tobi Lütke, Shopify, who credits Harry Brundage

Anatomy of a slop grenade.



YOU

Generate it.



YOU

Skip reading it.

"Slop grenade": Tobi Lütke, Shopify, who credits Harry Brundage

Anatomy of a slop grenade.



YOU

Generate it.



YOU

Skip reading it.



A TEAMMATE

Hand it on.

"Slop grenade": Tobi Lütke, Shopify, who credits Harry Brundage

Anatomy of a slop grenade.



YOU	YOU	A TEAMMATE	SOMEONE ELSE
Generate it.	Skip reading it.	Hand it on.	**Someone else pays.**

"Slop grenade": Tobi Lütke, Shopify, who credits Harry Brundage

Anatomy of a slop grenade.



YOU	YOU	A TEAMMATE	SOMEONE ELSE
Generate it.	Skip reading it.	Hand it on.	**Someone else pays.**

Seconds to make. Hours to clean up.

"Slop grenade": Tobi Lütke, Shopify, who credits Harry Brundage

AI slop?

We had human slop first.

“Hey, Jen, can you review this for me?”

Click.

Done!



Defeats the purpose.

"The danger lies in how easily developers may accept AI-suggested code without validating it."

Are we becoming…the TSA?

https://doctorow.medium.com/https-pluralistic-net-2025-12-05-pop-that-bubble-u-washington-8b6b75abc28e

“Automation blindness.”

Rare things are hard to find.



Maybe we've become…
too reliant on AI?



https://bsky.app/profile/yallchemist.bsky.social/post/3mwvtuvkjq22t

AXIOS

Josephine Walker • SEP 03, 2025

Widespread AI outage underway



Illustration: Brendan Lynch/Axios

OpenAI's ChatGPT, Anthropic's Claude, and xAI's Grok are all down Thursday morning, according to the outage company Downdetector.

Why it matters: Outages are common, but it's unusual for all three to go down at once, especially as the world increasingly relies on AI models.

- There have also been reported issues with Google's Gemini, but the company has not issued a confirmed outage notice.

Driving the news: Anthropic's Claude dashboard is experiencing outages across Opus 4.8 and Opus 5, and says all other models "have recovered to baseline error rate."

- ChatGPT and Codex are actively experiencing "elevated errors," according to OpenAI's dashboard.
- Grok is "experiencing issues," according to xAI, which says it's working on the problem.



Andrej Karpathy
@karpathy

A few random notes from claude coding quite a bit last few weeks.

[...]

Atrophy. I've already noticed that I am slowly starting to atrophy my ability to write code manually. Generation (writing code) and discrimination (reading code) are different capabilities in the brain. Largely due to all the little mostly syntactic details involved in programming, you can review code just fine even if you struggle to write it.

Slopacolypse. I am bracing for 2026 as the year of the slopacolypse across all of github, substack, arxiv, X/instagram, and generally all digital media. We're also going to see a lot more AI hype productivity theater (is that even possible?), on the side of actual, real improvements.

[...]

3:25 PM · Jan 26, 2026

40.7K 1.6K replies 5.6K reposts

x.com/karpathy/status/2015883857489522876

The skill AI erodes first
is debugging.

AI and learning.

17%

lower quiz scores with AI

52 developers learned a new library.

Half used an AI assistant.

The biggest gap: debugging.

Shen and Tamkin, "How AI Impacts Skill Formation" (arXiv:2601.20245), 2026

An intern directing an intern.

A junior developer plus an agent. Who checks the work?

“Even as experts become the only people who can effectively check the work of ever more capable AIs, we are in danger of stopping the pipeline that creates experts.”

–Ethan Mollick, *Co-Intelligence*

You can’t review what you can’t read.

So yes. You still need to learn this.

Ever notice how AI is really good at the things you don’t know?

Two codebases, same assistant, completely opposite reaction.

Trust what you know,
verify what you don’t.

Today, the job is knowing what good looks like.

You still need to learn new things.

The reps rule.

Never written it?

Write it once yourself.

A new framework. A new concept.
A bug you don't understand.

Written it a thousand times?

Hand it off.

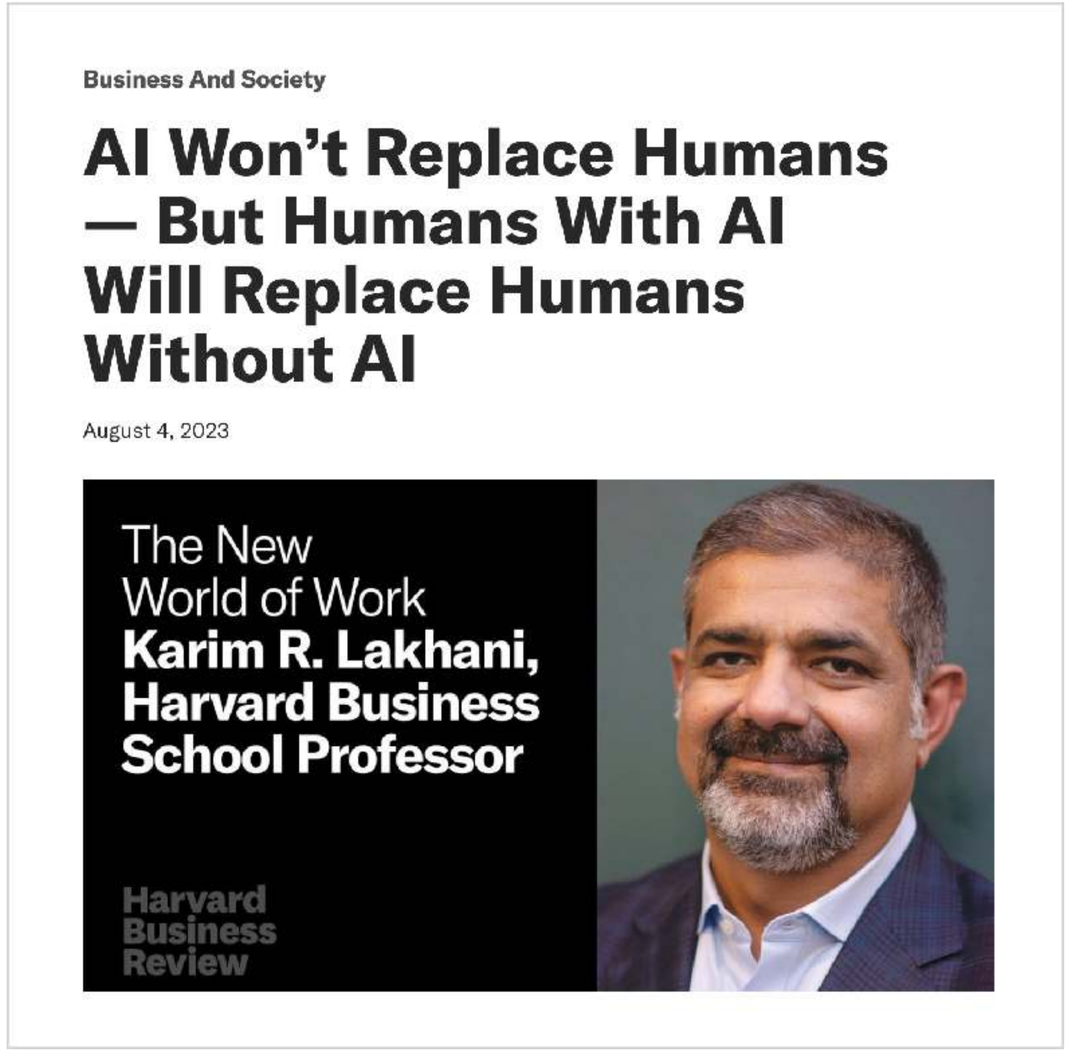
The thousandth controller. Boilerplate.
Things you've done many times.

“AI won’t replace humans. But humans with AI will replace humans without AI.”

Karim R. Lakhani

Professor, Harvard Business School

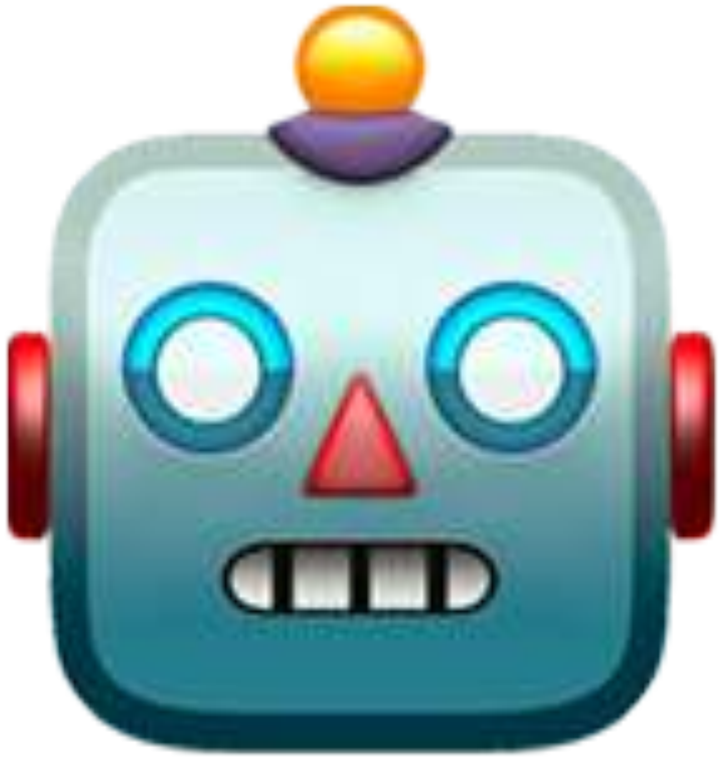
Harvard Business Review, August 4, 2023



Do the first ones by hand.
Watch the machine do the next thousand.

Stephen Wolfram on learning integrals, as told by Karim Lakhani

You can't send a robot
to the gym for you.



Know one level of abstraction below the one you use.

Know what's under the hood of the nail gun.

In the era of AI, you will spend even more time reading code.

Smells are your
review checklist now.

Know what good looks like?
Write it down.

Diagrams don't compile.
Neither do specs.

Tests do.



Ryan Dahl

@rough__sea · **Follow**

This has been said a thousand times before, but allow me to add my own voice: the era of humans writing code is over. Disturbing for those of us who identify as SWEs, but no less true. That's not to say SWEs don't have work to do, but writing syntax directly is not it.

11:02 AM · Jan 19, 2026

19.7K **Reply** **Copy link to post**

Read 931 replies

x.com/rough__sea/status/2013280952370573666



Ryan Dahl
@rough__sea

I want to share how I have developed celld - because it's a bit different and it's working well for me. You might have noticed that the http://github.com/denoland/celld repo is pretty bare - this is because it's an export of a much larger private repo: denoland/celld.dev

This larger repo contains: an llm-wiki with an enormous amount of design docs, a large test suite, a spec written in TLA+, the code for the website, scripts for spinning up fleets of celld instances on various clouds and running tests against them, and a speculative centralized managed control plane that I might someday release as a commercial product.

[...]

11:26 PM · Aug 24, 2026

1.1K 51 replies 73 reposts

x.com/rough__sea/status/2092091242377265562

The fix was a gate.

Not a better model.

Done means.

It has a test.

The tests pass.

You saw it work.

You said what you checked, and what you didn't.



daniel:// stenberg://
@bagder@mastodon.social

I hope I'm not the only one who think that a solid test suite for your project has proven more important than ever.

That's the wall crappy AI-generated PRs can't climb over.

Feb 03, 2026, 05:20 PM · · Web

99 boosts · **1** quote · **200** favorites

mastodon.social/@bagder/116009025260090059

Relationships matter.



Crawl. Walk. Run.

Start slowly.

The fundamentals didn't change.

The ratio did.

Writing code is now cheap.
Owning code is still expensive.

Owning code is the craft.
And the craft isn’t going away.



The Photographer.

Centro Português de Fotografia, Porto · Carlos Teixidor Cadenas, CC BY-SA 4.0

So now what?

Many of us enjoy the craft of building software.

Technology changes.

Constantly.

Our industry is evolving.

Some will be tempted to just…



It’s a bold strategy Cotton.
Let’s see if it pays off.

Two possible mindsets.

You can define yourself by what you've done in the past.

Or…

You can define yourself by the problems you will solve in the future.

You can be reactive.

Or proactive.

Whatever path you choose, change is inevitable.

You are responsible
for your career!

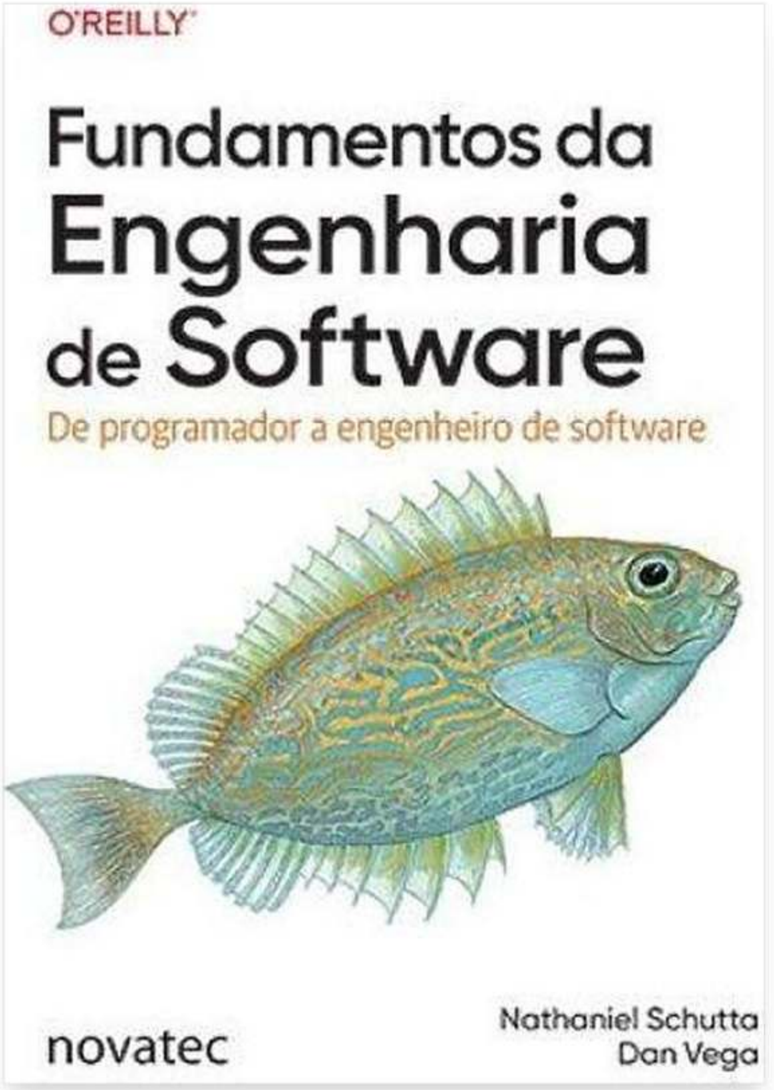
No one can do it for you.

You can't stop learning.

Fundamentals will
always serve you well.

“Excellence is doing ordinary things extraordinarily well.”

John W. Gardner



GIVEAWAY · 5 COPIES

Fundamentos da Engenharia de Software.

The Portuguese edition. Five copies, courtesy of O'Reilly and Novatec.

How to win: ??

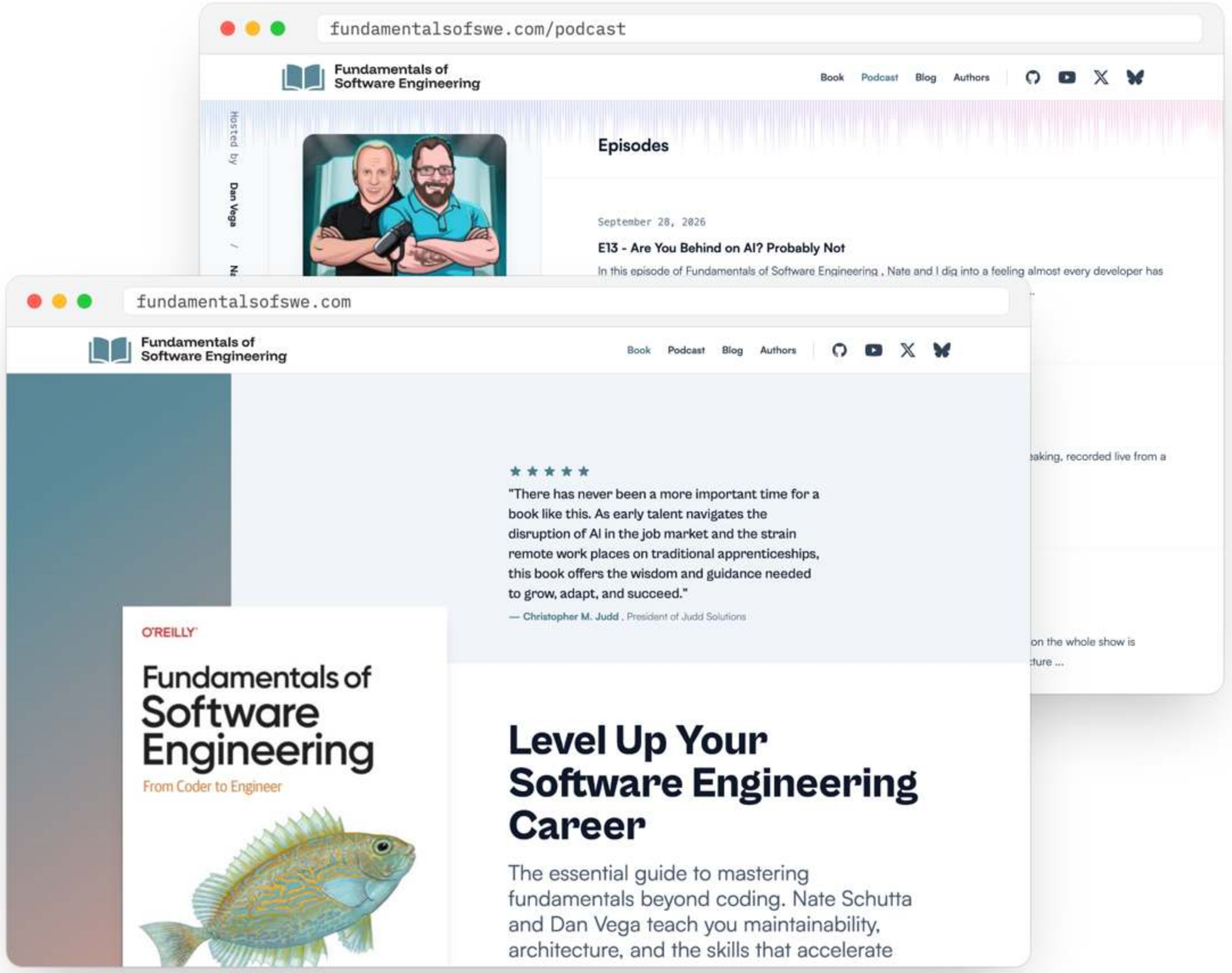
Learn more.

The book and our podcast.

fundamentalsofswe.com

LATEST EPISODE

Are you behind on AI? Probably not.



Brigado!

